FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Advanced)-2019

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

  Section-A (01– 07, 19 – 25, 37 - 43) *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 mark*** *for wrong answer*
  Partial Marks **+1** for each correct option provided no incorrect options is selected.

  Section-A (08 – 13, 26 – 31, 44 - 49) *contains 18 questions. Each of 2 Tables with 3 Columns and 4 Rows has three questions. Column 1 will be with 4 rows designated (I), (II), (III) and (IV). Column 2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column 3 will be with 4 rows designated (P), (Q), (R) and (S).*
  Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-C (14 – 18, 32 – 36, 50 - 54) *contains 15 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

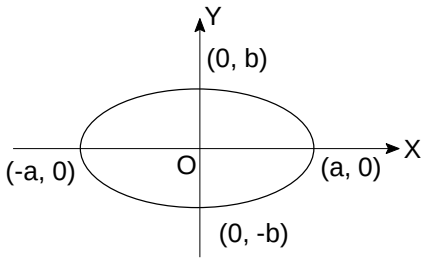
Physics

PART – I

SECTION – A
(More Than One Correct Type)

This section contains **7 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

1. Consider an elliptic disc having uniform distribution of mass M as shown in figure.
The moment of inertia (M.I.) about the major and minor axis are $\frac{1}{4}mb^2$ *and* $\frac{1}{4}ma^2$ respectively. Then



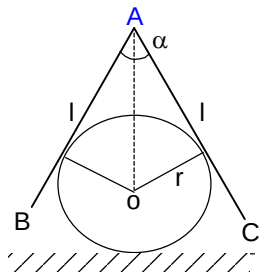
(A) The M.I. about a line, in its plane, passing through its centre and inclined at an angle θ to the major axis is $\frac{1}{4}m\left(b^2\cos^2\theta + a^2\sin^2\theta\right)$

(B) The M.I. about a chord of length 2r passing through O and lying in plane of elliptic disc is $\frac{1}{4}\frac{Ma^2b^2}{r^2}$

(C) The M.I. about a line perpendicular to the plane of elliptic disc and passing through the centre O is $\frac{1}{4}M\left(a^2+b^2\right)$

(D) The M.I. about a line, in its plane, passing through its centre and inclined at an angle θ to the minor axis is $\frac{1}{4}m\left(b^2\sin^2\theta + a^2\cos^2\theta\right)$

2. Two equal uniform rods of length l are joined at one end so that the angle between them is α and they rest in a vertical plane on a smooth sphere of radius r which is fixed to the ground.



(A) The rods are in stable equilibrium if $l > 4r\,\mathrm{cosec}\,\alpha$

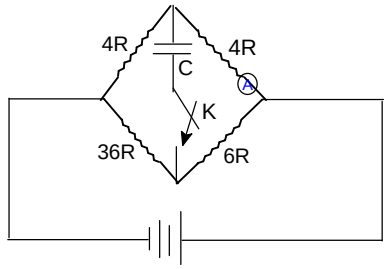
(B) The rods are in stable equilibrium if $l < 4r\,\mathrm{cosec}\,\alpha$

(C) The rods are in unstable equilibrium if $l > 4r\,\mathrm{cosec}\,\alpha$

(D) The rods are in unstable equilibrium if $l < 4r\,\mathrm{cosec}\,\alpha$

Space for Rough work

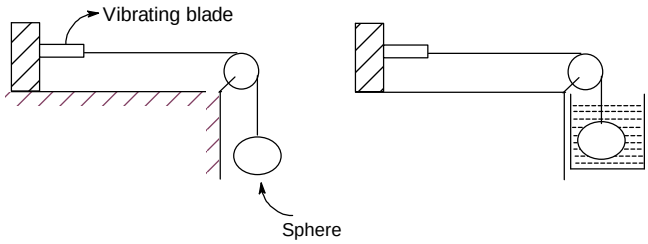
3. In the circuit shown below, the capacitor is initially uncharged. Immediately after the key K is closed, the reading of the ammeter is I.



(A) The Emf of battery is 8 IR
(B) The Emf of battery is 10 IR
(C) The steady state charge on capacitor is $\frac{25}{7}CIR$
(D) The steady state current through 6 R is $\frac{4}{21}I$

4. Which of the following is correct for Fission Nuclear Reactor?
(A) Slower neutrons are more effective in causing fission than faster neutrons in case of ^{235}U.
(B) Cadmium rod increase the reactor power when they are inserted and decrease when they are pulled back
(C) The kinetic energy of fast neutrons decreases when passed through heavy water
(D) Cadmium is a very good neutron absorber

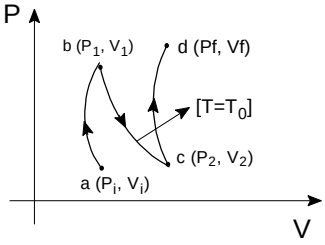
5. A string is tied at one end to a vibrating blade, and the other end is passes over a fixed pulley as shown in figure. A sphere of mass m hangs on the other end of the string. The string is vibrating in its n_1 harmonic. A container of water is raised under the sphere so that the sphere is completely submerged. In this configuration, the string vibrates in its n_2 harmonic.



(A) The magnitude of buoyant force is $mg\left[1-\left(\frac{n_1}{n_2}\right)^2\right]$
(B) The magnitude of buoyant force is $mg\left[1-\left(\frac{n_2}{n_1}\right)^2\right]$
(C) The radius of the sphere is $\left\{\frac{3m}{4\pi\rho_{water}}\left[1-\left(\frac{n_1}{n_2}\right)^2\right]\right\}^{1/3}$
(D) The radius of the sphere is $\left\{\frac{3m}{4\pi\rho_{water}}\left[1-\left(\frac{n_2}{n_1}\right)^2\right]\right\}^{1/3}$

Space for Rough work

6. One mole of an ideal monochromatic gas in an initial state a with pressure P_i and Volume V_i is to be taken to a final state d with $P_f = B^2 P_i$ *and* $V_f = V_i / B$ through the path $a \to b \to c \to d$ as shown in figure below for a particular values of $B(>1)$. Here $a \to b$ *and* $c \to d$ are adiabatic paths while $b \to c$ is an isothermal with temperature T_0. States b and c correspond to (P_1, V_1) *and* (P_2, V_2) respectively.



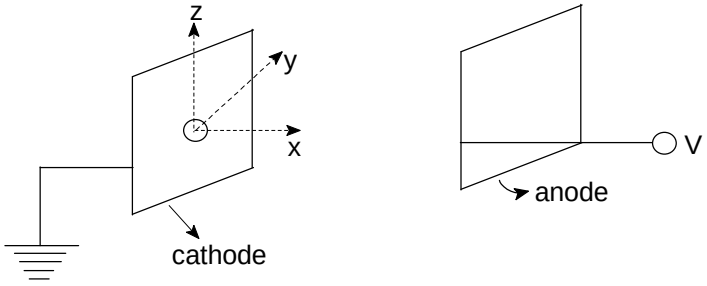
(A) The ratio of $\frac{V_2}{V_1}$ *is* $\sqrt{B}$

(B) The ratio of $\frac{V_2}{V_1}$ *is* $\sqrt{B^3}$

(C) Work done by the gas $\frac{RT_0}{2}\ln B + \frac{3}{2}P_i V_i (1-B)$

(D) Work done by the gas $\frac{RT_0}{2}\ln B + \frac{3}{2}(RT_0 - BP_i V_i)$

7. An electron initially at rest at point O (lying on the cathode) is acted upon by a magnetic field $\vec{B} = B_0 \hat{k}$ and electric field $\vec{E} = E_0 \hat{i}$. This electric field results due to the potential difference V between the cathode and anode separated by a distance d as shown in figure. Dimension of anode & cathode plates are very large compared to 'd'.



(A) The electron will not reach the anode if $B_0 = \frac{\pi}{2}\sqrt{\frac{mE_0}{2qd}}$

(B) The electron will not reach the anode if $B_0 = \frac{\pi}{2}\sqrt{\frac{mE_0}{qd}}$

(C) The electron will just reach the anode in time $t = \frac{m\pi}{2qB}$

(D) The electron will just reach the anode in time $t = \sqrt{\frac{2md}{qE}}$

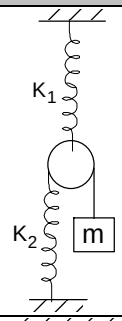
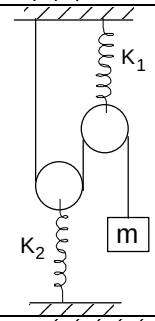
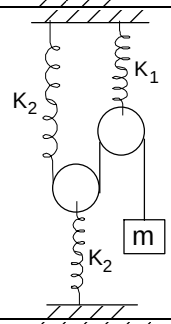
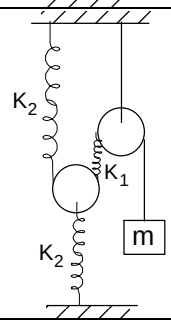
Space for Rough work

(Matching Type)
(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer 8, 9 and 10 by appropriately matching the information given in the three columns of the following table.

The Time period of spring-mass system executing SHM is given by $T = 2\pi\sqrt{\frac{M}{K_{eff}}}$. Column–1 shows the spring mass system executing SHM. Column–2 shows the force constant (spring constant) of combination while column–3 represents the time period of the oscillation.

Column–1		Column–2		Column–3	
(I)		(i)	$\frac{1}{K}=\frac{4}{K_1}+\frac{4}{K_2}$	(P)	$T=2\pi\sqrt{4M\left(\frac{1}{K_1}+\frac{1}{K_2}\right)}$
(II)		(ii)	$\frac{1}{K}=\frac{4}{K_1}+\frac{1}{K_2}$	(Q)	$T=2\pi\sqrt{M\left(\frac{4}{K_1}+\frac{1}{K_2}\right)}$
(III)		(iii)	$\frac{1}{K}=\frac{1}{K_1}+\frac{5}{K_2}$	(R)	$T=2\pi\sqrt{M\left(\frac{4}{K_1}+\frac{5}{K_2}\right)}$
(IV)		(iv)	$\frac{1}{K}=\frac{4}{K_1}+\frac{5}{K_2}$	(S)	$T=2\pi\sqrt{M\left(\frac{1}{K_1}+\frac{5}{K_2}\right)}$

8. If the time period of the combination of spring mass-system shown in column I

(III) $\left[if\ k_1 = k_2 = K\right]$ *is* $2\pi\sqrt{\frac{6M}{K}}$] The correct matching for the system is

(A) (III) (iii) (S) (B) (IV) (iii) (S)
(C) (I) (iii) (S) (D) (II) (iii) (S)

9. If spring constant for the combination shown in the column I (II) becomes twice if K_1 & K_2 becomes twice. Then correct matching shown for the spring-mass system is

(A) (I) (i) (P) (B) (II) (i) (P)
(C) (II) (i) (Q) (D) (I) (ii) (Q)

10. Which of the following is correct matching?

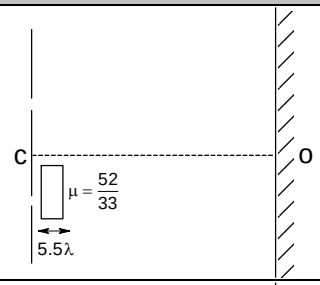
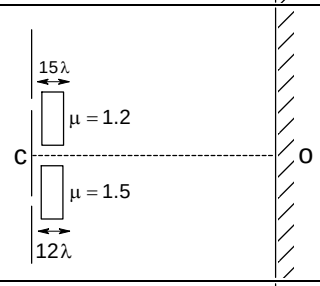
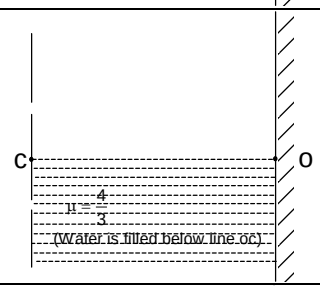
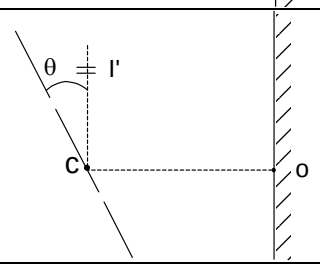
(A) (I) (i) (P) (B) (II) (ii) (Q)
(C) (III) (iv) (R) (D) (III) (iii) (S)

Space for Rough work

Answer 11, 12 and 13 by appropriately matching the information given in the three columns of the following table.

Every figure in column 1 shows a YDSE Apparatus with two slits of equal width emitting coherent light of equal intensity I and wavelength 'λ' in phase. Distance between midpoint of slits and midpoint of screen (which are in same horizontal line) is $D = 10^7\,\lambda$. The distance between slits is $d = \frac{8.82}{\pi} \times 10^4\,\lambda$. The plane containing slits and the plane of screen are vertical except for figure in column 1, Row 4 where, the plane of slits is tilted at an angle of $1'\left(= \frac{1^\circ}{60}\right)$ with the vertical but the plane of screen is here also vertical.

Column 2 represents the intensity of light.
Column 3 depicts [n] where [.] is a greatest integer function and 'n' represents number of fringes shifted through 'O' when the apparatus are formed with a standard YDSE apparatus. A standard YDSE apparatus is the one shown in column 1, row 4, with plane of slits vertical or $\theta = 0^\circ$ as shown in figure.

Column–1		Column–2			Column–3
(I)		(i)	0	(P)	0
(II)		(ii)	I	(Q)	3
(III)		(iii)	3I	(R)	8
(IV)		(iv)	4I	(S)	33, 33, 333

11. For setup in Column 1, Row 4, which of the following options from Column 2 & 3 are correct match?
(A) (IV) (ii) (Q) (B) (IV) (iii) (Q)
(C) (IV) (iii) (R) (D) (IV) (ii) (R)

12. Which of the following is a correct match?
(A) (I) (ii) (R) (B) (III) (ii) (S)
(C) (III) (iii) (P) (D) (I) (iii) (R)

13. For setup in Column 1, Row 2, which of the following options from Column 2 & 3 are correct match
(A) (II) (i) (Q) (B) (II) (iv) (P)
(C) (II) (i) (P) (D) (II) (iv) (Q)

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

14. If we model the electron as a uniform sphere of radius r_e spinning uniformly about an axis passing through its centre with angular momentum $L_e = 2\hbar$ and demand that the velocity of rotation at the equator cannot exceed the velocity c of light in vacuum. Then the minimum value of r_e *is* $\frac{n\hbar}{mc}$. Find n?

15. At normal incidence, a beam of light propagating in vacuum reflects off an interface with a medium of refractive index n=2.0. The fraction of energy reflected R is given by $R = \left(\frac{n-1}{n+1}\right)^2$ If the fractional error in the value of n is 3%, the fractional error in the estimation of R is.

16. A light beam from a laser pointer, on normal incidence creates a circular spot of diameter $2\times10^{-3}\,m$ on a perfectly reflecting surface. If the radiation pressure P on the surface due to totally reflected beam is $\frac{2}{3}\times10^{-5}\,N/m^2$, the time averaged power of laser beam (in mW) is nπ. Find n?

17. The electric field due to unknown charge distribution is given by $\vec{E} = \frac{k}{r^2}\exp(-4r)\hat{r}$, where k is a constant. The total charge in overall space is equal to

18. A spherical planet of radius R has a uniform density ρ and does not rotate. If the planet is made up of some liquid, the pressure at any point r from the centre is $\frac{2n\pi g\rho^2}{9}\left(R^2 - r^2\right)$. Find n?

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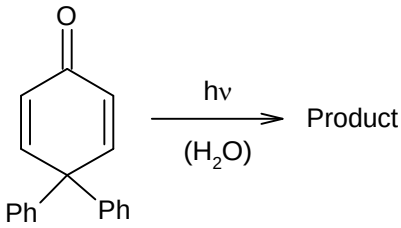
Chemistry

PART – II

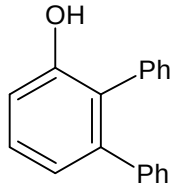
SECTION – A
(More Than One Correct Type)

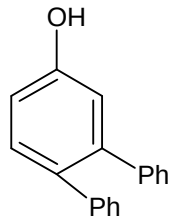
This section contains **7 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

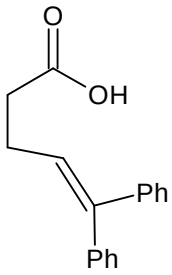
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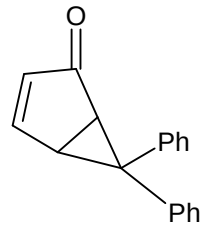


The product is/are

(A) 

(B) 

(C) 

(D) 

20. Which of the following is/are optically active?

(A) $\left[Co(en)_3\right]^{3+}$ (B) $\left[Co(ox)_3\right]^{3-}$

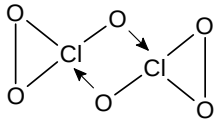
(C) cis $-\left[CoCl_2(en)_2\right]^{+}$ (D) trans $-\left[CoCl_2(en)_2\right]^{+}$

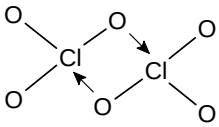
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21. Which is/are true about the N – N bond length among the following speices?
1. $H_2N - NH_2$ 2. N_2 3. $HN^+ - NH_2$ 4. N_2O
(A) N – N bond length is shortest in 2
(B) N – N bond length in 1 is shorter than that in 3
(C) N – N bond length in 3 is shorter than that in 1
(D) N – N bond length in 4 is intermediate between 1 and 3

22. By which of the following structure may Cl_2O_6 be represented?

(A) $ClO_2^+ClO_4^-$

(B) 

(C) 

(D) $ClO_3^-ClO_3^+$

23. Consider the following statements. Pick the correct one.
(A) All mono saccharides are reducing sugar
(B) Sucrose is a disaccharide and is a non – reducing sugar
(C) A solution consisting of equimolar glucose and fructose is dextrorotatory
(D) A solution consisting of equimolar sucrose and glucose in water dextrorotatory

24. Which of the following statements is/are correct?
(A) First dissociation constant of maleic acid is higher than that of first dissociation constant of fumaric acid
(B) Second dissociation constant of fumaric acid is higher than that of second dissociation constant of maleic acid
(C) pK_a value of salicylic acid is higher than that of benzoic acid
(D) pK_b of aniline is lower than that of o - toluidine

25. The $Al(OH)_3$ is involved in the following two equilibria,

$$Al(OH)_3(s) \rightleftharpoons Al^{3+}(aq.) + 3OH^-(aq.); K_{sp}$$

$$Al(OH)_3(s) + OH^-(aq.) \rightleftharpoons Al(OH)_4^-(aq); K_c$$

Which of the following relationship is correct at which solubility is minimum?

(A) $\left[OH^-\right] = \left(\frac{K_{sp}}{K_c}\right)^{\frac{1}{3}}$

(B) $\left[OH^-\right] = \left(\frac{K_c}{K_{sp}}\right)^{\frac{1}{4}}$

(C) $\left[OH^-\right] = \sqrt{\left(\frac{K_{sp}}{K_c}\right)^{\frac{1}{4}}}$

(D) None of these

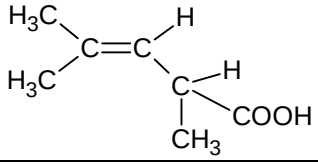
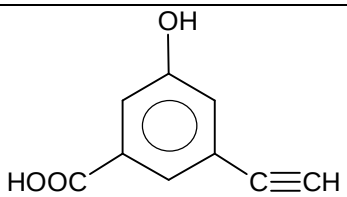
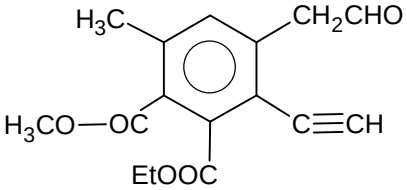
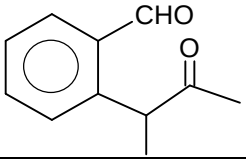
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(Matching Type)
(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer 26, 27 and 28 by appropriately matching the information given in the three columns of the following table.

Match the following

Column – 1		Column – 2		Column – 3	
(I)		(i)	Compound show optical activity	(P)	1 mol CH_3MgX is consumed
(II)		(ii)	CO_2 liberated with $NaHCO_3$	(Q)	2 mols CH_3MgX is consumed
(III)		(iii)	Pink colour of $KMnO_4$ is decolorized	(R)	3 mols CH_3MgX is consumed
(IV)		(iv)	Given precipitate with Tollen's reagent	(S)	5 mols CH_3MgX is consumed

26. Correct combination is

(A) $(II)(iii)(Q)$ (B) $(I)(iv)(P)$

(C) $(III)(iv)(S)$ (D) $(IV)(i)(R)$

27. Incorrect combination is

(A) $(I)(i)(P)$ (B) $(II)(iv)(R)$

(C) $(III)(ii)(Q)$ (D) $(III)(iv)(P)$

Space for Rough work

28. Correct combination is

(A) $(I)(iii)(Q)$ (B) $(II)(ii)(S)$

(C) $(IV)(iv)(P)$ (D) $(I)(ii)(P)$

Answer 29, 30 and 31 by appropriately matching the information given in the three columns of the following table.

Match the following

Column – 1		Column – 2		Column – 3	
(I)	$\underset{(1\,mol)}{P_4(s)} + NaOH + H_2O \longrightarrow$	(i)	Disproportionation reaction	(P)	An atom in the product is in the highest oxidation no.
(II)	$\underset{(1\,mol)}{FeS_2(s)} + O_2 \xrightarrow{\Delta}$	(ii)	$\frac{3}{5}$ mol of gas is formed in the product	(Q)	An atom in the product is in the lowest oxidation no.
(III)	$\underset{(1\,mol)}{H_3PO_3} \xrightarrow{\Delta}$	(iii)	2 mol of gas is formed in the product	(R)	An atom in the product has zero oxidation no.
(IV)	$ClO_3^- + \underset{(1\,mol)}{Cl^-} + H^+ \longrightarrow$	(iv)	1 mol of gas is formed in the product	(S)	One of the product in water behaves as an acid

29. Which of the following is the only correct combination?

(A) $(I)(iv)(R)$ (B) $(II)(i)(S)$

(C) $(IV)(ii)(R)$ (D) $(III)(iii)(P)$

30. Which of the following is the only INCORRECT combination?

(A) $(III)(i)(P)$ (B) $(III)(iv)(Q)$

(C) $(II)(iii)(S)$ (D) $(I)(iv)(Q)$

31. Which of the following is the correct combinations?

(A) $(I)(i)(Q)$ (B) $(II)(iv)(P)$

(C) $(III)(ii)(R)$ (D) $(IV)(iii)(S)$

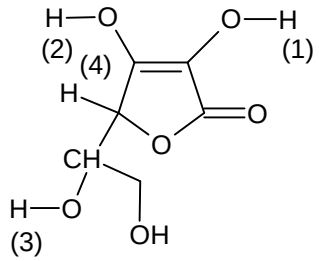
Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

32. 1 gm of complex $\left[Cr(H_2O)_5\,Cl\right]Cl_2.H_2O$ (Mol.Wt. $= 266.5$) was passed through a cation exchanger to produce HCl. The acid liberated was diluted to 1 litre. The normality of acid solution is 7.5×10^{-x} N. The value of x is

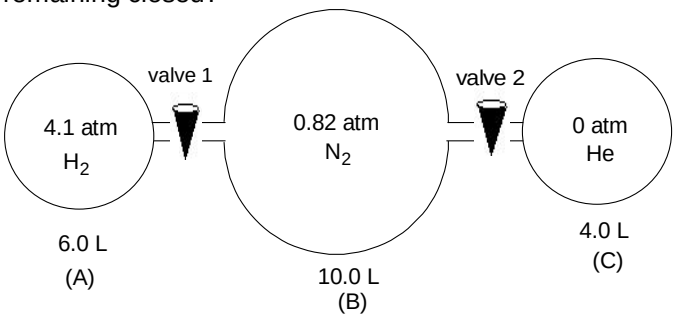
33. The density of steam at 27^0C and 8.314×10^4 pascal is 0.8 kg/m^3. The compressibility factor would be $\frac{a}{b}$. The value of a + b = ?

34. 

Identify the most acidic H in ascorbic acid (vitamin C)

35. Dark gray material (A) when it is heated at high temperature in air gives gas (B). Gas (B) is colourless and odourless which can not decolourize acidified $K_2Cr_2O_7$ solution. One of the allotropic forms of (A) is (A'), however (A) is the thermodynamically more stable than (A'). (A) on oxidation by hot concentration HNO_3 gives an acid (C), whereas (A) is unaffected by concentrated acids.
How many COOH groups are present in one molecules of acid (C)?

36. The figure given below shows three glass chambers that are connected by valves of negligible volume. At the outset of an experiment, the valves are closed and the chambers contain the gases as detailed in the diagram. All the chambers are at the temperature of 30 K and external pressure of 1.0 atm. What will be the work done by N_2 gas when valve -2 is opened and valve -1 remaining closed? What will be the work done by N_2 gas when valve – 2 is opened and valve -1 remaining closed?



Space for Rough work

Mathematics

PART – III

SECTION – A
(More Than One Correct Type)

This section contains **7 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

37. Let $f(x)=\frac{x^2-x+1}{x^2+x+1}$ *and* $g(x)=2f^3(x)-9f^2(x)+12f(x)+5$. In which of the following interval(s) is g(x) increasing?

(A) $(-2,-1)$ (B) $\left(-1,-\frac{1}{2}\right)$

(C) $\left(-\frac{2}{5},0\right)$ (D) $(1,\infty)$

38. Let $f(x)=\int_0^{\sin^2 x}\sin^{-1}\sqrt{t}\,dt+\int_0^{\cos^2 x}\cos^{-1}\sqrt{t}\,dt,\ x\in\left(0,\frac{\pi}{2}\right)$. Then the value of f(0) equals:-

(A) $f(1)$ (B) $2f\left(\frac{1}{2}\right)$

(C) $f\left(\frac{1}{2}\right)$ (D) $2f(1)$

39. Let S be set of all value (s) of 'a' which the inequality $x^2+|x-a|-3<0$ is satisfied by at least one negative 'x'. Which of the following can be subset of S?

(A) $(-3,2)$ (B) $(0,3)$

(C) $(-4,1)$ (D) $(-2,3)$

40. The vector $\vec{a}=-\hat{i}+2\hat{j}+\hat{k}$ is rotated through right angle such that $\vec{a}$ crosses through the Y- axis, while undergoing rotation. In the new position $\vec{a}$ is given by:-

(A) $\sqrt{2}\left(\hat{i}+\hat{j}-\hat{k}\right)$ (B) $\sqrt{2}\left(-\hat{i}-\hat{j}+\hat{k}\right)$

(C) $\sqrt{3}\left(\hat{i}+\hat{k}\right)$ (D) $\sqrt{3}\left(-\hat{i}-\hat{k}\right)$

Space for Rough work

41. Consider the lines $L_1 : x = y = z$ *and* $L_2 : 2x + y + z - 1 = 0 = 3x + y + 2z - 2$. Which of the following is (are) correct?

(A) The shortest distance between L_1 *and* L_2 *is* $\sqrt{2}$

(B) The shortest distance between L_1 *and* L_2 *is* $1/\sqrt{2}$

(C) The plane containing L_2 and parallel to L_1 *is* $y - z + 2 = 0$

(D) The plane containing L_2 and parallel to L_1 *is* $x - y + 1 = 0$

42. Let P(A) and P(B) be the probabilities of occurrence of two events A and B respectively. If $P(A) = \frac{3}{5}$ *and* $P(B) = \frac{2}{3}$ then which of the following is (are) correct?

(A) $P\left(A \cap \overline{B}\right) \le \frac{1}{3}$ (B) $P(A \cup B) \ge \frac{2}{3}$

(C) $\frac{4}{15} \le P(A \cap B) \le \frac{3}{5}$ (D) $P\left(\frac{\overline{A}}{B}\right) \le \frac{3}{5}$

43. If $z_1 = a + ib$ *and* $z_2 = c + id$ are two complex numbers such that $|z_1| = |z_2| = 1$ *and* $\text{Re}\left(z_1 \overline{z}_2\right) = 0, (a, b, c, d \in R)$. Then for the pair of complex numbers $\omega_1 = a + ic$ *and* $\omega_2 = b + id$, which of the following is (are) correct?

(A) $|\omega_1| = 1$ (B) $\text{Re}\left(\omega_1, \overline{\omega}_2\right) = 0$

(C) $\text{Re}\left(\overline{\omega}_1, \omega_2\right) = 1$ (D) $\text{Re}\left(\omega_1, \overline{\omega}_2\right) = 1$

Space for Rough work

(Matching Type)
(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer 44, 45 and 46 by appropriately matching the information given in the three columns of the following table.

Answer Questions 8 -10 by appropriate matching the information given in the three columns of the following table.

Column 1, 2, 3 contains equations of hyperbola, co-ordinates of a point & value of 'a'

Column – 1		Column – 2		Column – 3	
(I)	$x^2 - y^2 = 4$	(i)	$(a-1, a-2)$	(P)	$\frac{5}{2}$
(II)	$xy = 4$	(ii)	$(1-a, a)$	(Q)	$\frac{1}{6}$
(III)	$\frac{x^2}{4} - y^2 = 1$	(iii)	$(2a, a-1)$	(R)	2
(IV)	$x^2 - \frac{y^2}{4} = 1$	(iv)	(4a, a)	(S)	0

44. The tangents to suitable hyperbola (column I) from the point (column II) are on both branches for the value of a (column III). Which of the following options is the only correct combination.
(A) (I) (ii) (S) (B) (I) (iii) (S)
(C) (II) (iv) (R) (D) (IV) (i) (R)

45. Suitable point (column II) lies on director circle of hyperbola (column I) for the value of 'a'. Which of the following options is the only correct combination.
(A) (I) (ii) (P) (B) (II) (i) (R)
(C) (II) (iv) (S) (D) (IV) (iii) (Q)

46. Suitable point (column II) lies on hyperbola (column I) for the value of 'a' (column III). Which of the following options is the only correct combination
(A) (III) (i) (R) (B) (II) (iv) (P)
(C) (I) (iii) (Q) (D) (IV) (i) (R)

Space for Rough work

Answer 47, 48 and 49 by appropriately matching the information given in the three columns of the following table.

Column 1, 2, 3 contain Trigonometric Function, Maximum Value and Minimum Value respectively					
Column – 1		**Column – 2**		**Column – 3**	
(I)	$f(x) = \frac{7 + 6\tan x - \tan^2 x}{1 + \tan^2 x},\ x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	(i)	$\frac{1}{4}$	(P)	1
(II)	$f(x) = 4\cos x + 3\cos\left(x + \frac{\pi}{2}\right) + 3,\ x \in R$	(ii)	1	(Q)	-2
(III)	$f(x) = \sin^6 x + \cos^6 x,\ x \in R$	(iii)	8	(R)	$-\frac{1}{4}$
(IV)	$f(x) = \cos x.\cos\left(x + \frac{\pi}{3}\right).\cos\left(x - \frac{\pi}{3}\right),\ x \in R$	(iv)	2	(S)	$\frac{1}{4}$

47. Which of the following option is the only CORRECT combination?
(A) (I) (iii) (Q) (B) (II) (iii) (P)
(C) (III) (ii) (P) (D) (IV) (ii) (R)

48. Which of the following option is the only CORRECT combination?
(A) (I) (iii) (P) (B) (II) (iii) (Q)
(C) (III) (iv) (P) (D) (IV) (i) (Q)

49. Which of the following option is the only CORRECT combination?
(A) (I) (iii) (S) (B) (II) (iii) (R)
(C) (III) (iv) (S) (D) (IV) (i) (R)

Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. Consider two circles $C_1 : x^2 + y^2 + 4x + 4y - 1 = 0$ *and* $C_2 : x^2 + y^2 + 6x + 2y - 7 = 0$. The line $L_1 : x + 2y + 3 = 0$ intersects C_1 at points A and B and the line $L_2 : 2x + 3y + \lambda = 0$ intersects C_2 at point C and D. The value of $(2\lambda - 1)$ such that points A, B, C and D are concyclic, is:

51. If M is a 3×3 matrix, such that $\det(M) = 2$ *and* $M^T M = I$, (M^T is the transpose of matrix M). Then, the value of $\det(M - I)$ is:

52. The coefficient of x^{70} in the product $(x-1)(x^2-2)(x^3-3).....(x^{12}-12)$ is:

53. If 'z' is a complex number satisfying the equation $11z^{10} + 10iz^9 + 10iz - 11 = 0$, then the value of $3|z|$ is:

54. The largest integer value of 'k' for which the equation $e^x - 1 - k\tan^{-1}x = 0$, possess only one real root, is:

Space for Rough work